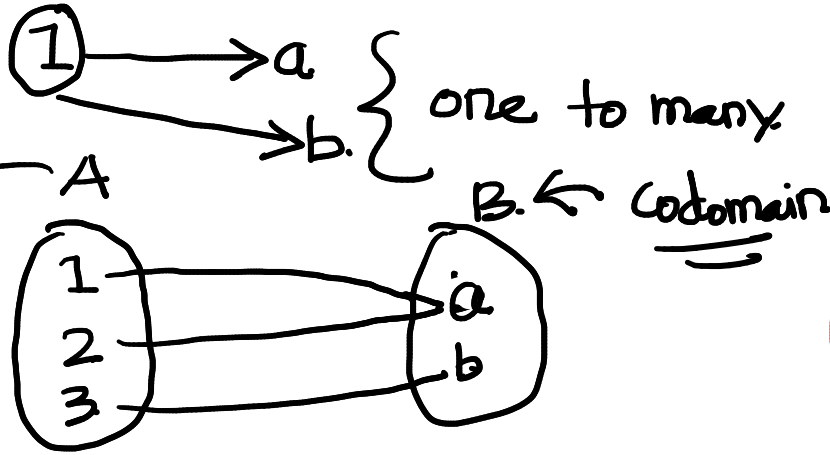
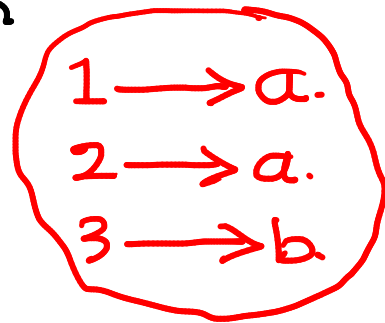


Function is relation which is not one-many.

$$\{(1, a), (1, b), (2, a), (3, b)\} \subset \underline{A \times B}$$



1 $\neq$ 2 but f(1) = f(2)



$A = \{1, 2\}$ $B = \{a, b\}$

$\underline{R} = \{(1, a), (2, a)\}$ - This a function from "A" to "B"

$\not\subset \underline{A \times B}$

$\Rightarrow R$ is not a relation.

$\Rightarrow R$ is not function.

one-one functions

If no "two elements" in the domain of f .

Correspond to the same element in the codomain of f .



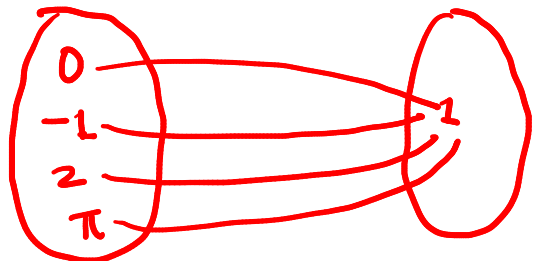
for any $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$

$\star \checkmark f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

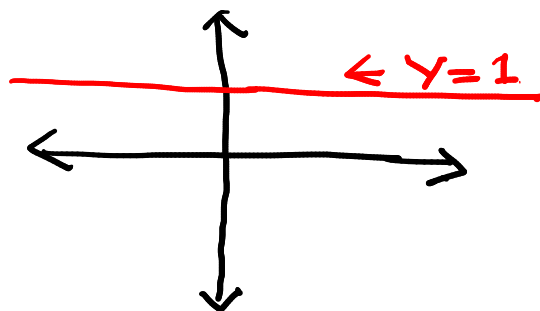
Ex: $f: \mathbb{R} \rightarrow \mathbb{R}$.

$y = f(x) = 1 \quad \forall x \in \mathbb{R}$
 $\hookrightarrow$ Constant

$\mathbb{R}$.



- This is not one-one.



Ex: $f: \mathbb{R} \rightarrow \mathbb{R}$

$x \mapsto |x|$

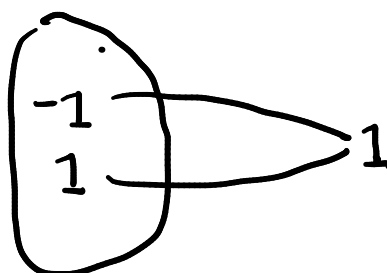
$\begin{cases} -1 \end{cases} \rightarrow 1$

$\begin{cases} 1 \end{cases} \rightarrow 1$

$2 \rightarrow 2$

$-2 \rightarrow 2$

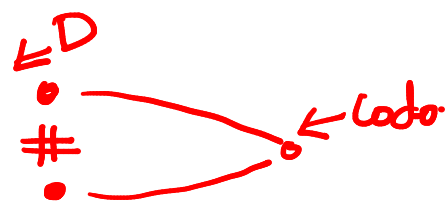
$\leftarrow$ one-one.



Ex: $\mathbb{R} \rightarrow \mathbb{R}$
 $f(x) = 2x$?

one-to-one

for any $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$



$\checkmark \underline{x, y} \in \text{Dom}(f) \text{ s.t. } x \neq y \Rightarrow 2x \neq 2y \leftarrow f \text{ is one-one.}$
 $\Rightarrow f(x) \neq f(y)$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto x^2$$

$$\begin{matrix} -1 \\ 1 \end{matrix} \mapsto 1$$

$$(-\infty, \infty) \rightarrow \mathbb{R}$$

$$x^5 + 7x^2 + 2x + 19$$

Roots

Even: $\Rightarrow$ not one-to-one.

odd: x^3 - One-one

Sink - not one-one

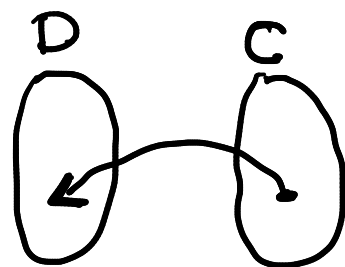
$$\underline{x \neq y} \Rightarrow \underline{x^2 \neq y^2}$$

WLog

$$x > y \geq 0 \Rightarrow x^2 > y^2 \geq 0$$

$$f(x) > f(y)$$

$$f(x) \neq f(y)$$



Range = Codomain

Greatest Int. f. // Floor function.

$\lfloor x \rfloor$:= The largest integer that is less than or equal to " x ".

$$\lfloor 2.3 \rfloor = 2.$$

$$\lfloor 3 \rfloor = 3.$$

$$\lfloor -1.1 \rfloor = -2.$$

$$\lfloor -3 \rfloor = -3.$$

$$\lfloor -3.3 \rfloor = -4.$$

$$2 \leq 2.3.$$

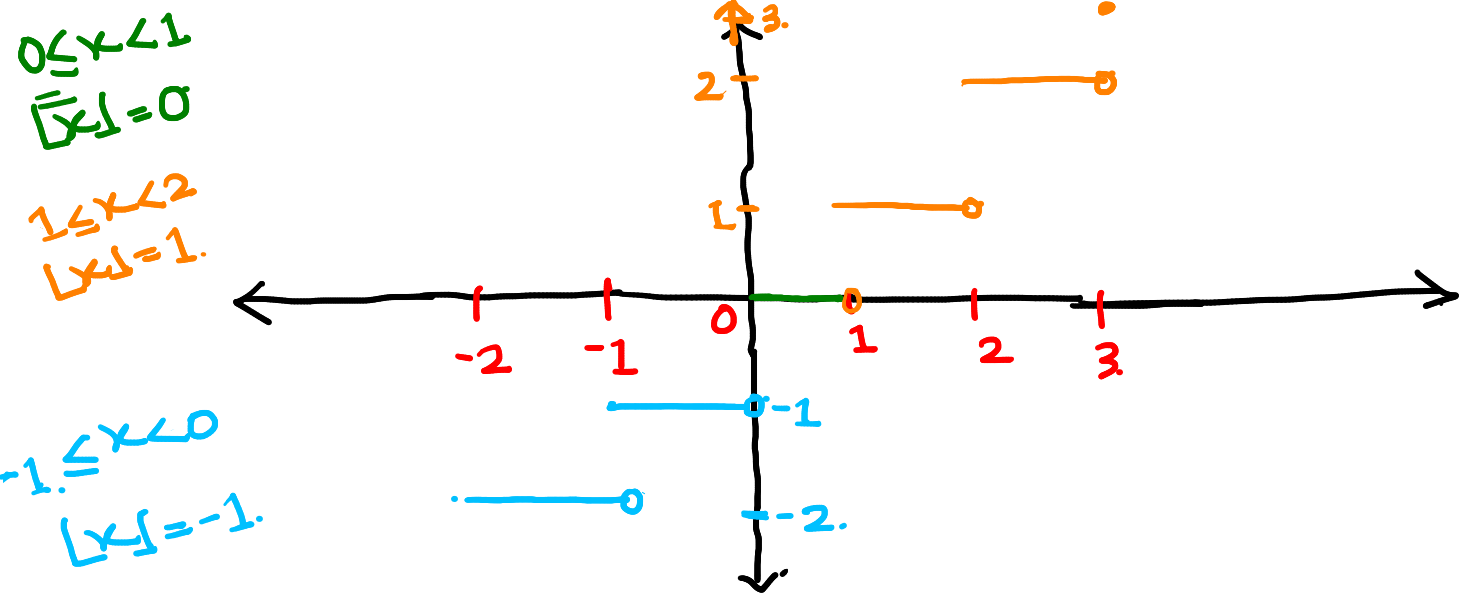
$$-- 0, 1, \underline{2}, 2.3, \underline{3}, \underline{4}, --$$

$$.. -3, \underline{-2}, -1.1, \underline{-1}, 0, --$$

$f: \mathbb{R} \rightarrow \mathbb{R}$. one-to-one?

$$x \mapsto \lfloor x \rfloor$$

$$\begin{cases} 2 \mapsto 2 \\ 2.3 \mapsto \lfloor 2.3 \rfloor = 2 \end{cases}$$



Smallest Int.

$\lceil x \rceil =$ Smallest Integer greater than or equal to " x "

$\lceil 2.1 \rceil = 3$ -- 1, 2, 2.1, 3, 4, --

$\lceil 2 \rceil = 2$

$\lceil -1.1 \rceil = -1$ -- -2, -1.1, -1, 0, --

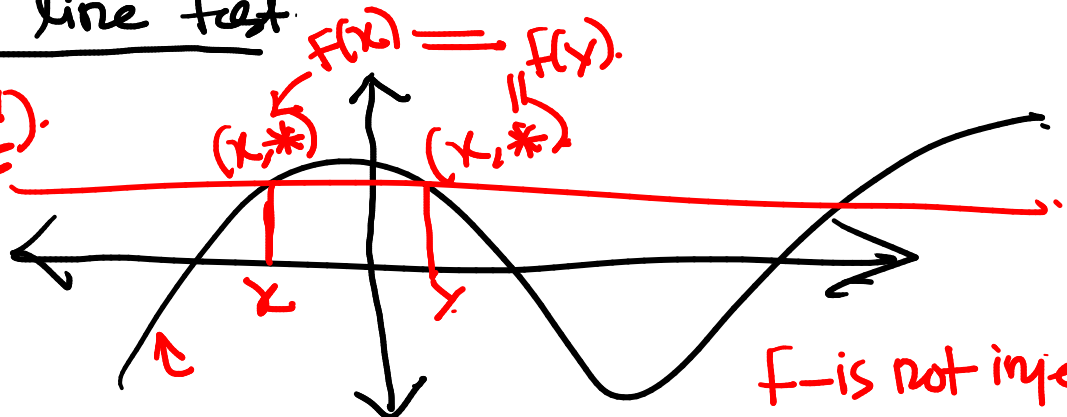
$\lceil -1 \rceil = -1$

— This is not one-to-one
 = Draw the graph.

Horizontal line test

But $f(x) = f(y)$.

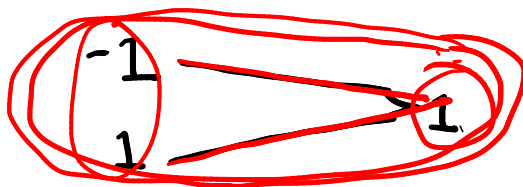
$x \neq y$



Invertible function:

$$f: \mathbb{R} \rightarrow [0, \infty)$$

$$x \mapsto x^2 \quad \leftarrow \text{not one-one}$$



$\left\{ \begin{array}{l} g(1) = -1 \\ g(1) = 1 \end{array} \right. \quad \leftarrow \text{not a function}$

Class of one-one function.

If $f: A \rightarrow B$ (on-to), then $f^{-1}: B \rightarrow A$

Such that 1. $f \circ f^{-1} = I$

$$I(x) = x$$

$$f^{-1}: B \rightarrow A$$

$$(f \circ f^{-1})(x) = I(x) = x \quad \forall x \in B$$



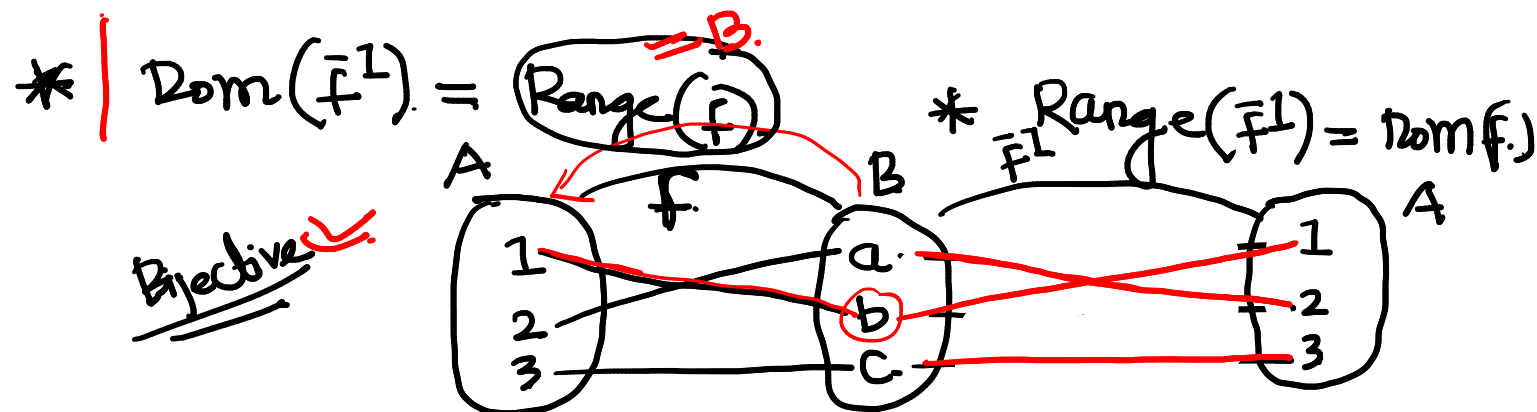
$$f(f^{-1}(x))$$

$$I: B \rightarrow B$$

$$f \circ f^{-1}: B \rightarrow B$$

2. $f^{-1} \circ f = I$

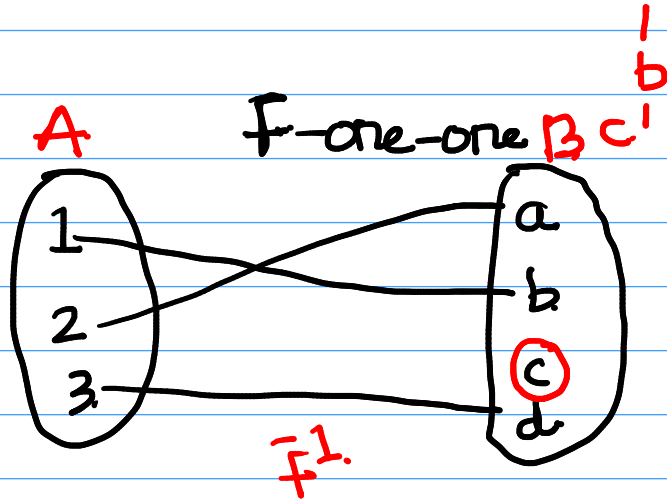
$$(f^{-1} \circ f)(x) = x \quad \forall x \in A$$



$$\bullet \quad \underline{\underline{f^{-1} \circ f = I_A}} \quad (\underline{\underline{f^{-1} \circ f}})(1) = \underline{\underline{f^{-1}(b)}} = 1$$

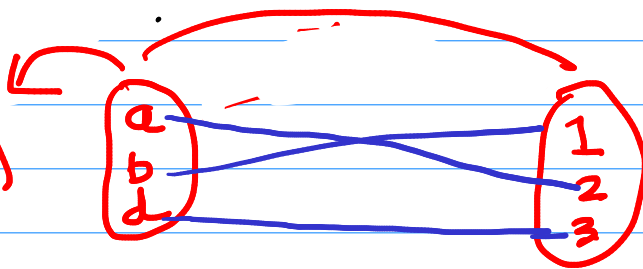
2 = 2

$$\bullet \quad f \circ \underline{\underline{f^{-1}}} = I_B \quad (f \circ \underline{\underline{f^{-1}}})(a) = f(2) = a$$



$$\text{Dom}(\underline{\underline{f^{-1}}}) = \text{Image}(f) = \{a, b, d\}$$

$$\text{Dom}(\underline{\underline{f^{-1}}}) = \text{Range}(f) \neq B$$



$$f \circ \underline{\underline{f^{-1}}} = I_{\{a, b, d\}}$$

$$\underline{\underline{f^{-1} \circ f = I_A}}$$

$$f: [0, \infty) \rightarrow [0, \infty)$$

$$\underline{\underline{f(x) = x^2 + 1}}$$

one-one — ✓

one-to — ✗

$$[1, \infty)$$

↑ Image(f)

$$x^2 \geq 0 \Rightarrow x^2 + 1 \geq 1$$

$\mathbb{R} \rightarrow \mathbb{R}$

$$f: [0, \infty) \rightarrow [1, \infty) \quad (\text{Both one-one \& on-to.})$$

$$\underline{\underline{f^{-1}: [1, \infty) \rightarrow [0, \infty)}}$$

$$f \circ \underline{\underline{f^{-1}}} = I_{[1, \infty)}$$

$$\underline{\underline{f^{-1} \circ f}}$$

$$\Rightarrow (f \circ \underline{\underline{f^{-1}}})(x) = I_{[1, \infty)}(x) = x \quad \forall x \in [1, \infty)$$

$$y = f^{-1}(x)$$

$$\Rightarrow f(\underline{f^{-1}(x)}) = x \quad \forall x \in [1, \infty)$$

$$\Rightarrow \underline{f(y)} = x$$

$$\Rightarrow y^2 + 1 = x$$

$$\Rightarrow y^2 = x - 1$$

$$\Rightarrow y = \pm \sqrt{x-1}$$

$$\Rightarrow \boxed{f^{-1}(x) = \pm \sqrt{x-1}}$$

~~$$f^{-1}(x) = -\sqrt{x-1} \in \text{1-ve values}$$~~

$$f^{-1}(x) = \sqrt{x-1}$$

$$(f \circ g)(x) = \frac{2x}{3-x}$$

$$0 \notin \text{Dom}(g)$$

$$3 \notin \text{Dom}(f \circ g)$$

$$f(x) = \frac{2}{x-1} \quad g(x) = \frac{3}{x}$$

$$\mathbb{R} \setminus \{3\}$$

$$\text{Dom}(f \circ g)$$

$$0 \notin \text{Dom}(f \circ g)$$

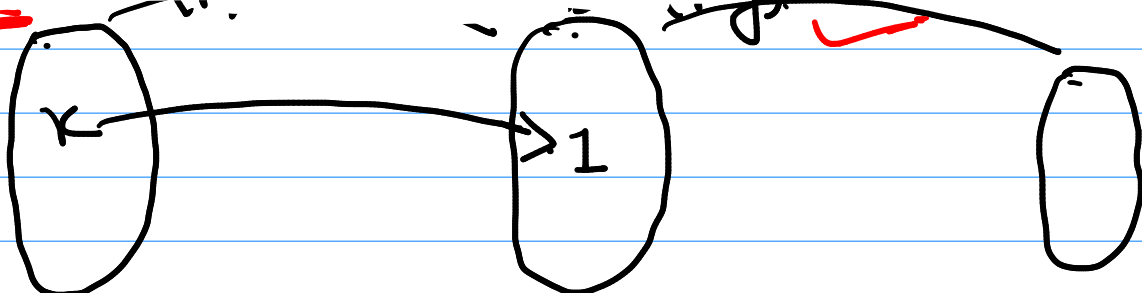
$$\boxed{\mathbb{R} \setminus \{3, 0\}}$$

$$x \in \text{Dom}(g) \Rightarrow x \in \text{Dom}(f \circ g)$$

$$\{x \mid g(x) \in \text{Dom}(f)\} - \text{must not inside}$$

$$1 \in \text{Dom}(f)$$

$$(f \circ g)(x) = \underline{f(1)}$$



$$y = 50$$

✓. $x \notin \text{Dom}(f) \Rightarrow x \notin \text{Dom}(f \circ g) \leftarrow \mathbb{R} \setminus \{0, 1, 3\}$

$\text{Dom}(f) = \mathbb{R} \setminus \{1, 2, 3\}$

(X): $\underline{g(x) = 1} \quad \parallel \quad \begin{matrix} 3/x \\ g(x) = 2 \\ \underline{3/2} \end{matrix}$

$\Rightarrow 3/x = 1$

$\Rightarrow x = 3$

$g(x) = 3$
(1)

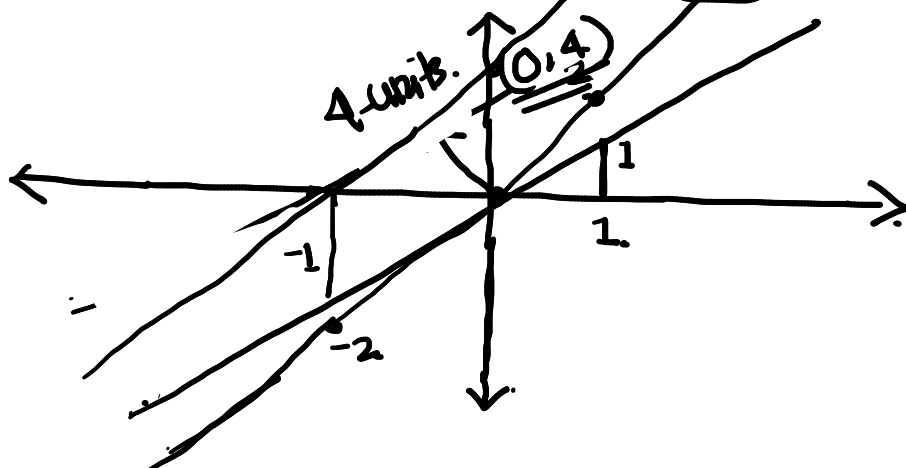
$\{x \mid g(x) \notin \text{Dom}(f)\} = \{3\}$

Graph:

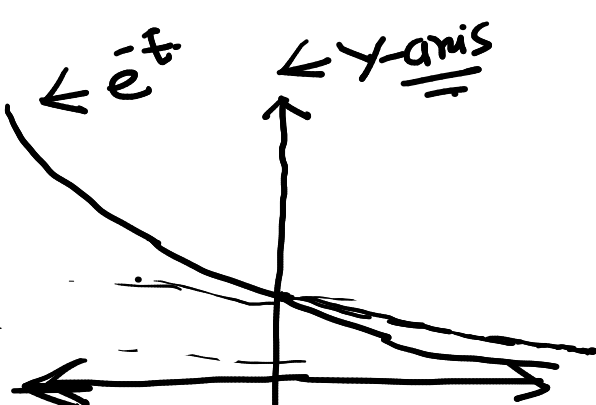
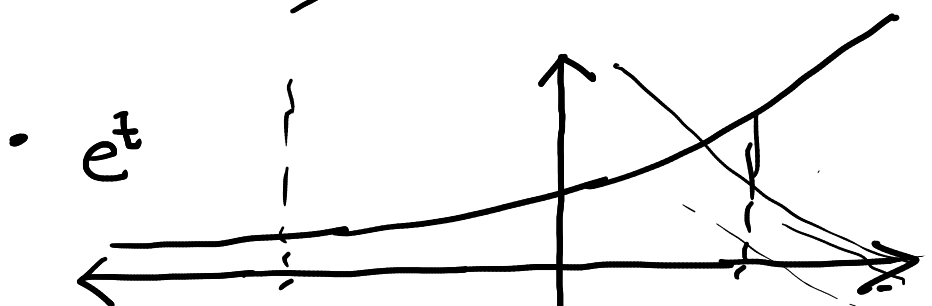
$R(t) = 50 - 100e^{-0.2t}$

$F(x) = x$

$f(x) = 2x + 4$



$e^{-0.2t} = e^{0.2(-t)}$



$\frac{0.02(-t)}{e}$

e^{-t}

at $t = -100$

$t = 100$

$e^2 = 1/e^2$

e^{100}

$t = 100$

$e^{-100} = 1/e^{100}$